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TOP HAT STRUCTURE FOR ISOLATION CAPACITORS

Abstract

An isolation capacitor structure reduces the likelihood of breakdown in the passivation layers by physically re-shaping or dividing the top plate of the isolation capacitor into two segments. In that way, the electric field is driven down and away from the passivation surfaces. One embodiment utilizes a series capacitor formed by the top metal plate of the capacitor and an additional “top hat” plate above the top metal plate that redirects the fields into the main isolation capacitor. Vias may be included between the top hat plate and the top metal plate. Another approach reshapes the top plate to have an integrated top hat structure and achieves similar results of directing charge down and away from the passivation layer surface breakdown paths.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation of U.S. patent application Ser. No. 16/728,224, filed Dec. 27, 2019, which is hereby incorporated by reference in its entirety herein.

BACKGROUND

Field of the Invention

[0002] This application relates to capacitors and more particularly to capacitors used in isolation applications.

Description of the Related Art

[0003] Isolation communication channels are utilized in various applications where necessary to prevent current flow between separate isolated electric circuits while still maintaining communication between the circuits. The isolation may be required for signal isolation, safety, or for other reasons. The isolation communication channels may be implemented using capacitive techniques.

[0004] Referring to FIG. 1 a typical isolation capacitor **100** includes a top metal plate **101**, a bottom metal plate **103** above a silicon substrate **105**, and a region of insulator material such as SiO₂ (intermetal dielectric (IMD)) **106** between the bottom metal plate and the top metal plate. Passivation layers **107**, which vary from fab to fab and process to process, and are typically comprised of multiple stacked materials such as SiO₂ and SiN, serve as a protective barrier to protect the integrated circuit. The isolation capacitor **100** allows for low voltage signaling, e.g., of approximately 1-3 volts for transfer of information (e.g., based on pulse width of the signal or other information in the signal). The bond wire **109** used in the low voltage signaling can experience large non-signal voltages from hundreds to even thousands of volts relative to the integrated circuit in which the capacitor is disposed. Those high voltages are also applied to the top plate through the bond wire. The IMD layer **106** isolates the large non-signal voltages from other circuits in the integrated circuit.

[0005] A guard ring **110** is placed at a distance from the isolation capacitor to protect circuits outside of the guard ring. The guard ring is formed by metal in each metal layer coupled by vias to form a vertical guard ring coupled to ground so any charge received at the guard ring goes to ground. The guard ring is placed, e.g., at a distance 10 times the thickness of the intermetal dielectric layer (IMD), which is the distance between the top metal plate and the bottom metal plate. Note that a horizontal capacitor is formed between the top metal layer and the guard ring. Electric fields **121** under large voltages that can be experienced by the isolation capacitor extend into the passivation layers and can cause the passivation layers to break down, catastrophically damaging the integrated circuit. Accordingly, improved isolation techniques are desirable.

SUMMARY OF EMBODIMENTS OF THE INVENTION

[0006] In one embodiment an integrated circuit includes a first conductive plate above a substrate and a second conductive plate above the first conductive plate. A first dielectric layer is between the first conductive plate and the second conductive plate. A third conductive plate is above the second conductive plate and a second dielectric layer between the second conductive plate and the third conductive plate. The third conductive plate is smaller than the second conductive plate. In an

embodiment, a plurality of vias are between the third conductive plate and the second conductive plate to electrically connect the third conductive plate and the second conductive plate.

[0007] In another embodiment an integrated circuit includes a first conductive plate above a substrate, a second conductive plate above the first conductive plate, and a first dielectric layer between the first conductive plate and the second conductive plate. The second conductive plate is formed with a center portion having a greater thickness than outer portions of the second conductive plate.

[0008] In another embodiment a method of making an isolation capacitor includes forming a first conductive plate above a substrate, forming a first dielectric layer above the first conductive plate, forming a second conductive plate above the first dielectric layer, forming a second dielectric layer above the second conductive plate, and forming a third conductive plate above the second dielectric layer and the third conductive plate is smaller than the second conductive plate. The method may further include forming a plurality of vias connecting the second conductive plate and the third conductive plate.

[0009] In another embodiment, a method of forming an integrated circuit includes, forming a first conductive plate above a substrate, forming a first dielectric layer above the first conductive plate, and forming a second conductive plate above the first dielectric layer with a center portion of the second conductive plate having a greater thickness than an outer portion of the second conductive plate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The present invention may be better understood, and its numerous objects, features, and advantages made apparent to those skilled in the art by referencing the accompanying drawings.

[0011] FIG. 1 illustrates a prior art isolation capacitor and guard ring.

[0012] FIG. 2 illustrates a simplified model of electric fields in a cross section of a parallel plate capacitor.

[0013] FIG. 3 illustrates fringe fields extending from edges of a finite parallel plate capacitor.

[0014] FIG. 4 illustrates high fields concentrated at the edge of the top metal plate of an isolation capacitor can reach the passivation layers and create a breakdown path resulting in charge transfer from the top metal to the guard ring.

[0015] FIG. 5 illustrates that electric fields terminating at an edge of a top plate of an isolation capacitor can penetrate the passivation layers.

[0016] FIG. 6 illustrates an embodiment of an isolation capacitor structure having a “top hat” metal plate above the top metal plate of capacitor thereby reducing fields in the passivation layers.

[0017] FIG. 7 shows a top view of the top hat structure above the IMD layer 617.

[0018] FIG. 8 illustrates an embodiment having a “top hat” metal plate above the top metal plate of a capacitor with vias in the dielectric layer coupling the top hat metal plate to the top metal plate.

[0019] FIG. 9 illustrates another embodiment that reshapes the top metal plate to incorporate an integral top hat.

[0020] FIG. 10 illustrates another embodiment of isolation capacitor in which the top hat plate is placed off center of the top metal plate.

[0021] The use of the same reference symbols in different drawings indicates similar or identical items.

DETAILED DESCRIPTION

[0022] Isolation capacitor structures such as shown in FIG. 1 have a tendency to break down at voltages much lower than theoretical for an oxide based capacitive structure. That unexpected breakdown can occur in the passivation layers. The breakdown is believed to be the result of failure

along resistive paths at the surface (likely the result of charge traps created at interface layers during passivation material processing and between layers of passivation materials). Embodiments described herein reduce the likelihood of breakdown in the passivation layers either through physically re-shaping or dividing the top plate of the isolation capacitor into two segments. In that way, the electric field is driven down and away from the passivation surfaces. One embodiment utilizes a series capacitor formed by the top metal plate and an additional “top hat” plate above the top metal plate that actively redirects the fields into the main isolation capacitor while not appreciably reducing necessary dielectric thickness. A related, second approach uses vias between the top hat plate and the top metal plate. Another embodiment reshapes the top plate to have an integrated top hat and achieves similar results of directing charge down and away from the passivation layer surface breakdown paths.

[0023] Before proceeding with additional details of various embodiments, FIG. 2 illustrates simplified model of electric fields in a cross section of a parallel plate capacitor **200**. The parallel plate capacitor includes a top metal plate **201** and a bottom metal plate **203**. Field lines **205** extend through the insulating material from the top metal plate to the bottom metal plate. The conventional parallel plate capacitor equation is $C = \epsilon_{\text{sub.o}} \times k \times A / d$, where $\epsilon_{\text{sub.o}}$ is permittivity of free space, k is material relative permittivity (written sometimes as $\epsilon_{\text{sub.r}}$), A is plate area, and d is the spacing between the top plate **201** and the bottom plate **203**. The electric field $E = V / d$, where d is the spacing between plates and V is the applied voltage (or $E = k \times Q / r_{\text{sup.2}}$, where k is the electric constant of ($9 \times 10^9 \text{ Nm}^2 / \text{C}^2$), Q is source charge, and r is separation distance between charges in meters). The simple capacitor equation shows a uniform field between the plates as a result of an applied voltage. Evenly spaced field lines **205** represent the uniform E-field. However, this uniform field would only exist in an infinitely wide structure. In actuality, the electric field at the edges is higher, and at sharp metal corners, the field can be infinitely high.

[0024] Referring to FIG. 3, rather than a uniform electric field as shown in FIG. 2, a finite width parallel plate capacitor has fringe fields extending from the plate **301** to plate **303** as shown by the field arrows **305**. The fringe fields cause a field increase at plate corners as shown by the increased number of field lines that terminate near the corners of the capacitor plates. While theory says that an infinite field exists at a sharp corner, in reality, materials would breakdown in an infinite field.

[0025] Referring to FIG. 4, high fields concentrated at the edge of the top metal plate **401** can reach the passivation layers to create a breakdown path **402** resulting in charge transfer from the top metal **401** to the guard ring **410** along resistive paths at the surface of interfaces of the passivation layers or even at interfaces between in situ steps of the same layer. The more field seen at the weak leak in the passivation layers (e.g., at interfaces of different passivation layers), the more likely a breakdown will occur. Typical passivation layer materials used include high density plasma (HDP) oxide, plasma enhanced (PE) oxide, and SiN, and silicon rich oxide (SRO). The passivation layers are highly susceptible to process variation due to Gaussian distributions associated with deposition, chemical mechanical polishing (CMP) planarization steps, and other process steps associated with making the passivation layers. The particular materials used and thicknesses of the layers and thus the interfaces between layers vary significantly between fabrication facilities and even different generations of processes within the same fabrication facility. Thus, it would be desirable to reduce the likelihood of passivation layer failure without having to individually address process variations and variations in manufacturing approaches used to make isolation capacitors. Therefore embodiments draw down fields away from the passivation layers and thereby reduce the likelihood of failure in the passivation layers in an approach that is substantially immune to process variations.

[0026] FIG. 5 illustrates electric fields for a structure such as shown in FIG. 1 with a top metal plate **501** and a bottom metal plate **503**. The simulation shows that the field is concentrated at the edge of the top metal plate **501** and penetrates the passivation layers **507**. FIG. 5 shows contour lines **509**, **511**, and **515** that approximate the electric field of the simulation results. The field inside

contour line **509** has a maximum above $4\text{ kV}/\mu\text{m}$. The field between contour line **509** and **511** ranges between 1 (or more) $\text{kV}/\mu\text{m}$ and $0.8\text{ kV}/\mu\text{m}$. Finally, the field between field line **511** and **515** represents a field of between approximately $0.8\text{ kV}/\mu\text{m}$ to $0.6\text{ kV}/\mu\text{m}$. The field lines are approximate and will vary according to thicknesses of metal, types dielectric used, etc. Referring again to FIG. **1**, the top corner of the top metal plate is very close to the interfaces in the passivation layers that are most likely to experience failure. As described above, the electric field falls off with the square of the distance. Accordingly, embodiments described herein increase separation between high fields associated with the top metal plate and passivation interface layers, and thereby reduce the electric field seen in the passivation layers and thus reduce the likelihood of horizontal failure caused by current flow between the top metal plate and the guard ring.

[0027] Referring to FIG. **6**, an embodiment includes a metal “top hat” plate above the top metal layer. The isolation capacitor structure **600** shown in FIG. **6**, includes a top metal plate **601** and a bottom metal plate **603** separated by an IMD layer **606**. The IMD layer **606** is, e.g., 5-25 microns thick, with the thickness depending on such factors as the capacitance desired and the material used for the IMD layer. A new top metal top hat plate **615** sits above the top metal plate **601** and is separated from the top metal plate by IMD material **617** between the top metal plate **601** and the top hat structure **615**. The IMD material could be a standard oxide such as SiO_2 and be the same as the oxide in the stack of the primary isolator capacitor formed by top metal plate **601** and bottom metal plate **603**. In other embodiments the IMD layer **716** is an alternative insulator material such as SiN , SRO , or SiON . The structure shown in FIG. **6** provides a series capacitor with the added capacitor formed by the top hat plate and the top metal plate of the primary capacitor. Note that since the top hat structure is now the highest metal, it is referred to as being top metal (TM), whereas the top metal plate is now TM-1, or one metal layer down. For example, in an embodiment, the top hat is in M8 and the top metal plate is in M7.

[0028] In a preferred embodiment, the top hat structure is smaller than the top metal plate. For example, in one embodiment, the top hat plate **615** is formed as a $76\text{ }\mu\text{m}\times 76\text{ }\mu\text{m}$ plate while the top metal plate **601** is $120\text{ }\mu\text{m}\times 120\text{ }\mu\text{m}$. The size of the top hat plate should be large enough to support the bond wire **609**. The dielectric material between the top hat plate and the top metal plate should be thin enough to allow some limited voltage drop, thereby allowing charge to flow while minimizing current induced self-heating and thus damage to the dielectric layer. In various embodiments the thickness of the top hat metal **615**, the thickness of the top metal plate, and the thickness of the dielectric material between TM and TM-1 varies. For example, in an embodiment, the dielectric material **617** between the top hat plate and the top metal plate is $1.7\text{ }\mu\text{m}$, the top hat plate is $3\text{ }\mu\text{m}$, and the top metal plate is thinner than the top hat plate at $0.7\text{ }\mu\text{m}$. With 5000 volts applied between the top hat plate and the bottom plate, in simulation the voltage at the top plate drops to 2715 V and the fields were reduced in the passivation layer as compared to the fields illustrated in FIG. **5**. In other embodiments, the thickness of the dielectric material **617** is $0.5\text{ }\mu\text{m}$, the top hat plate is $1\text{ }\mu\text{m}$ and the top metal plate is $3\text{ }\mu\text{m}$. The metal plates can be formed of copper, aluminum, or any metal or metal alloy suitable in semiconductor processing. The dielectric material of the IMD layer **617** can of course be different than SiO_2 as described earlier. With SiO_2 being used for IMD **617**, the voltage at the top metal plate (TM-1) was 4097 V so the top series capacitor had a voltage drop of approximately 903 V. With SiN , the voltage at the top metal plate (TM-1) was 4420 V and thus the voltage drop across the top series capacitor was reduced as compared to using SiO_2 . With a thin top hat plate (e.g., $\sim 1\text{ }\mu\text{m}$) and a thick top metal plate (e.g., $\sim 3\text{ }\mu\text{m}$) and a thin dielectric layer **617**, the field in the passivation layer was significantly reduced such that fields below approximately $300\text{ V}/\mu\text{m}$ appear in the passivation layer in simulations. FIG. **7** shows a top view of the top hat structure **615** above the IMD layer **617**.

[0029] Making the series capacitor embodiment of FIG. **6** includes depositing the bottom metal plate **603**, depositing the IMD layer **606**, depositing the top metal plate **601**, the IMD layer **617**, and the top hat plate **615** along with appropriate patterning and etch steps for the various capacitor

components. By making the top hat plate smaller than the top metal plate, the edge of the top hat plate is farther from the guard ring. In addition, the top metal plate blocks the top hat plate from seeing the bottom metal plate resulting in no significant field being generated due to interactions between the bottom metal plate and the top hat plate. That reduces the field at the edge of the top hat plate as compared to embodiments without the top hat plate and thus reduces field in the passivation layer. Moving the top metal plate to TM-1 moves the high fields associated with the edge of the top metal plate further from the layers of the passivation layer that are more susceptible to failure. Simulation shows that utilizing the series capacitor structure with the top hat successfully reduced the field in the passivation layer but there was still significant field in the IMD layer **617**. Accordingly, there is some risk of break down in the IMD layer **617** although such a break down would not necessarily cause a catastrophic failure.

[0030] Referring to FIG. **8**, another embodiment of the isolation capacitor **800** includes a top metal plate **801** and a bottom metal plate **803** separated by an IMD layer **806**. A top metal top hat plate **815** sits on top of the top metal plate **801** separated by IMD material **817** between the top metal plate **801** and the top hat plate **815**. In embodiments, the IMD material is a standard oxide such as SiO₂ and similar to IMD layer **806** or could be an alternative insulator material such as SiN, SRO, or SiON. The isolation capacitor **800** includes conductive vias **819** formed in the dielectric layer **817** to electrically couple the top hat plate **815** and the top metal plate **801**. The addition of the conductive vias avoids damage to the dielectric of the top capacitor (formed by top hat plate **615** and top metal plate **601** in the embodiment of FIG. **6**) by ensuring that the top hat plate **815** and the top metal plate **801** are at the same voltage. The approach shown in FIG. **8** allows for a thicker top hat structure, which allows more passivation on top of the isolation capacitor **800**. A thicker top hat structure can include a thicker IMD layer **817** thereby shifting the top plate further down from the passivation layers subject to failure. The field lines terminate at the end of the top metal plate **801** (at TM-1) since the top hat plate **815** and the top metal plate **801** are at the same potential. In simulations applying 5000 V between the top hat plate and the bottom metal plate, the field in the passivation layers is reduced to 300 V/μm or less. Note that moving the top metal plate down can decrease the distance between the top metal plate **801** and the bottom metal plate **803**, therefore the capacitance and isolation capabilities of the isolation capacitor can be affected. So there is a tradeoff between moving the top metal plate closer to the bottom metal plate that should be considered.

[0031] In an embodiment, the top hat plate **815** is formed as a 76 μm×76 μm plate while the top metal plate **801** is 120 μm×120 μm. The size of the top hat plate should be large enough to support the bond wire **809**. The dielectric material between the top hat plate and the top metal plate no longer needs to allow for a voltage drop due to the presence of the vias allowing for thicker IMD **817**. In various embodiments the thickness of the top hat metal **815**, the thickness of the top metal plate **801**, and the thickness of the dielectric material **817** between TM and TM-1 varies. For example, in an embodiment, the dielectric material **817** between the top hat plate and the top metal plate is 2.0 μm, the top hat plate is 3 μm, and the top metal plate is thinner than the top hat plate at 0.7 μm. With 5000 volts applied between the top hat plate and the bottom plate, the fields were reduced in the passivation layer as compared to simulation of the embodiment illustrated in FIG. **5**. In other embodiments, the thickness of dielectric material **817** is 0.5 μm, the top metal plate is 3 μm and the top hat plate is 1 μm. The dielectric material of the IMD layer **817** can of course be different from SiO₂ as described earlier.

[0032] Referring to FIG. **9** in another embodiment isolation capacitor **900** reshapes the top metal plate **901** to increase the distance from the edge **920** of the top metal plate where the field lines terminate to the passivation layers susceptible to high electric fields from the top capacitor plate. The embodiment of FIG. **9** removes the insulator between the top metal plate and the top hat plate by forming an integral metal structure. The top metal plate **901** moves the top corner fields at corner **905** farther from the guard ring and moves the bottom corner fields at corners **902** and **904**

farther from interface layers. The top metal plate with an integrated top hat **921** can be formed either through a dual step metal deposition or deposition of a thicker top metal followed by a second pattern and etch to create the top hat shape (the raised region **921**). The top hat **921** should be sized to support the bond wire and in an embodiment has $76\text{ }\mu\text{m}\times 76\text{ }\mu\text{m}$ top surface while the top metal plate bottom surface is $120\text{ }\mu\text{m}\times 120\text{ }\mu\text{m}$. Thus, the top surface area of the top hat is less than 50% of the bottom surface area of the top metal plate. In a typical embodiment the top metal plate is $3\text{ }\mu\text{m}$ thick including of $2\text{ }\mu\text{m}$ for the top hat portion **921** and $1\text{ }\mu\text{m}$ for the extended (120×120) base. Other embodiments use different thicknesses according to process capabilities and design requirements.

[0033] FIG. **10** illustrates another embodiment of isolation capacitor **1000** in which the top hat plate **1015** is placed on the side of the top metal plate **1001** and the bottom plate **1003** is smaller than the top plate **1001**. The bottom plate **1003** can be outside the perimeter of the top hat plate **1015** and dielectric layer **1017** as shown or in another embodiment, part of the bottom plate extends under the top hat plate **1015** and dielectric layer **1017**.

[0034] Thus, various aspects have been described related to an isolation capacitor structure with a top hat plate that reduces the risk of breakdown at interfaces in the passivation layer. The description of the invention set forth herein is illustrative, and is not intended to limit the scope of the invention as set forth in the following claims. Other variations and modifications of the embodiments disclosed herein, may be made based on the description set forth herein, without departing from the scope of the invention as set forth in the following claims.

Claims

1-20. (canceled)

21. An integrated circuit comprising: a silicon substrate; a first conductive plate formed over the silicon substrate; a second conductive plate formed over a portion of the first conductive plate; a first dielectric region separating the first conductive plate and the second conductive plate, and no conductive path formed between the first conductive plate and the second conductive plate; a third conductive plate formed over a portion of the second conductive plate; a second dielectric region separating the second conductive plate and the third conductive plate, a thickness of the first dielectric region differing from a thickness of the second dielectric region, the third conductive plate being off center from the second conductive plate in a first lateral direction, the first conductive plate being off center from the second conductive plate in a second lateral direction opposite the first lateral direction, and the first conductive plate being at least partially outside a perimeter of the third conductive plate; and a plurality of conductive vias formed in the second dielectric region to electrically couple the second conductive plate and the third conductive plate.

22. The integrated circuit of claim 21 wherein the first conductive plate, the second conductive plate, and the third conductive plate are plates of an isolation capacitor.

23. The integrated circuit of claim 21 wherein a thickness of the third conductive plate differs from a thickness of the second conductive plate.

24. The integrated circuit of claim 21 wherein the thickness of the first dielectric region ranges between 5 microns and 25 microns.

25. The integrated circuit of claim 21 wherein the thickness of the second dielectric region ranges between 5 microns and 25 microns.

26. The integrated circuit of claim 21 further comprising a bond wire electrically coupled to the third conductive plate.

27. The integrated circuit of claim 26 wherein the third conductive plate is sized to support the bond wire.

28. The integrated circuit of claim 21 wherein the second dielectric region is formed of SiO_2 , SiN , silicon rich oxide, or SiON .

- 29.** The integrated circuit of claim 21 wherein a gap exists in a lateral plane between the first conductive plate and the third conductive plate.
- 30.** The integrated circuit of claim 21 further comprising a guard ring, the guard ring being further separated from the third conductive plate than the second conductive plate due to the third conductive plate being smaller than the second conductive plate, thereby reducing a likelihood of failure due to current flow between the third conductive plate and the guard ring.
- 31.** The integrated circuit of claim 21 wherein the first dielectric region is formed of a first dielectric material and the second dielectric region is formed of a second dielectric material different than the first dielectric material.
- 32.** An isolation capacitor comprising: a first conductive plate formed over a substrate, a second conductive plate formed over a portion of the first conductive plate, and a third conductive plate formed over a portion of the second conductive plate, the third conductive plate being off center from the second conductive plate in a first lateral direction, the first conductive plate being off center from the second conductive plate in a second lateral direction opposite the first lateral direction, and the first conductive plate being at least partially outside a perimeter of the third conductive plate; and a first dielectric region separating the first conductive plate and the second conductive plate with no conductive path formed between the first conductive plate and the second conductive plate, and a second dielectric region separating the second conductive plate and the third conductive plate, a top surface of the third conductive plate exposed by an opening in one or more passivation layers positioned above the second dielectric region, and a thickness of the first dielectric region differing from a thickness of the second dielectric region.
- 33.** The isolation capacitor of claim 32, further comprising a plurality of conductive vias formed in the second dielectric region to electrically couple the second conductive plate and the third conductive plate.
- 34.** The isolation capacitor of claim 32, wherein a thickness of the third conductive plate differs from a thickness of the second conductive plate.
- 35.** The isolation capacitor of claim 32, further comprising a bond wire electrically coupled to the third conductive plate, the third conductive plate sized to support the bond wire.
- 36.** The isolation capacitor of claim 32, wherein a gap exists in a lateral plane between the first conductive plate and the third conductive plate.
- 37.** The isolation capacitor of claim 32, further comprising a guard ring, the guard ring being further separated from the third conductive plate than the second conductive plate due to the third conductive plate being smaller than the second conductive plate, thereby reducing a likelihood of failure due to current flow between the third conductive plate and the guard ring.
- 38.** The isolation capacitor of claim 32 wherein the first dielectric region is formed of a first dielectric material and the second dielectric region is formed of a second dielectric material different than the first dielectric material.
- 39.** A method of manufacturing an isolation capacitor comprising: forming a first conductive plate of an isolation capacitor over a substrate and withing a first dielectric region; forming a second conductive plate of the isolation capacitor over a portion of the first dielectric region with no conductive path formed between the first conductive plate and the second conductive plate of the isolation capacitor, the first conductive plate and the second conductive plate separated by the first dielectric region; forming a second dielectric region over the second conductive plate; and forming a third conductive plate over a portion of the second dielectric region and such that the third conductive plate is off center from the second conductive plate in a first lateral direction, the first conductive plate being formed off center of the second conductive plate in a second lateral direction opposite the first lateral direction, and the first conductive plate being outside a perimeter of the third conductive plate.
- 40.** The method of claim 39 further comprising forming conductive vias in the second dielectric region connecting the second conductive plate and the third conductive plate.

